Class of November 10

Professor: Fco. Vazquez

1 Algebra skills

Simplify the following expresisons,

$$\begin{aligned}
&1)\quad x-[3a+2(-x+1)],\\
&2)\quad -[3x-2y+(x+2y)-2(x+y)-3(x+1)],\\
&3)\quad a-(x+y)-3(x-y)+2[-(x-2y)-2(-x-y)].
\end{aligned}$$

Find all real zeros of the following polynomials,

$$\begin{aligned}
&1)\quad P(x)=3x^3+5x^2-2x-4\\
&2)\quad P(x)=3x^4-5x^3-16x^2+7x+15\\
&3)\quad P(x)=x^4-6x^3+4x^2+15x+4\\
&4)\quad P(x)=3x^3+4x^2-x-2
\end{aligned}$$

2 Function analysis

Mention the domain and range of the following functions,

$$1)\quad f(x)=\sqrt{x^2-1},\qquad 2)\quad g(x)=\frac{2}{x(3x+5)},\qquad 3)\quad h(x)=-\frac{3}{2}x^2.$$

3 Modelling with math

a) A school fund-raising group sells chocolate bars to help finance a swimming pool for their physical education program. The group finds that when they set their price at x dollars per bar (where $0 < x \leq 5$), their total sales revenue (in dollars) is given by the function $R(x) = -500x^2 + 3000x$.

- Evaluate $R(2)$ and $R(4)$. What do these values represent?
- How many bars they need to sell to get even?

b) A right triangle has one leg twice as long as the other. Find a function that models its perimeter P in terms of the length x of the shorter leg.

c) A Norman window has the shape of a rectangle surmounted by a semicircle. A Norman window with perimeter 30 ft is to be constructed.

- Find a function that models the area of the window.
- Find the dimensions of the window that admits the greatest amount of light.